

Computer Architecture Lab Manual - Test Cases

Swap Two 8-bit Numbers

Test Case ID	Input / Scenario	Expected Result
TC01	A=12H, B=34H	Values swapped
TC02	A=00H, B=FFH	Values swapped correctly
TC03	A=55H, B=55H	No change after swap

Largest and Smallest of Two Numbers

Test Case ID	Input / Scenario	Expected Result
TC01	25H, 10H	25H largest, 10H smallest
TC02	01H, FFH	FFH largest
TC03	20H, 20H	Both equal

Positive, Negative or Zero Check

Test Case ID	Input / Scenario	Expected Result
TC01	05H	Display EE
TC02	80H	Display FF
TC03	00H	Display DD

4th Bit Check

Test Case ID	Input / Scenario	Expected Result
TC01	08H	Display FF
TC02	00H	Display DD
TC03	18H	Display FF

Sum of First Ten Natural Numbers

Test Case ID	Input / Scenario	Expected Result
TC01	1 to 10	Sum = 55
TC02	Correct iteration	Loop executes 10 times

TC03	Accumulator check	Stores final result
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Sum of Digits of 8-bit Number

Test Case ID	Input / Scenario	Expected Result
TC01	25H	Digit sum calculated
TC02	99H	Correct sum
TC03	00H	Result = 0

Reverse of 8-bit Number

Test Case ID	Input / Scenario	Expected Result
TC01	123	321
TC02	100	1
TC03	121	121

Palindrome Check

Test Case ID	Input / Scenario	Expected Result
TC01	121	Display FF
TC02	123	Display DD
TC03	11	Display FF

Odd or Even Check

Test Case ID	Input / Scenario	Expected Result
TC01	08H	Display DD
TC02	09H	Display FF
TC03	00H	Display DD

Count Number of Ones

Test Case ID	Input / Scenario	Expected Result
TC01	0FH	Count = 4
TC02	FFH	Count = 8
TC03	00H	Count = 0

Addition and Subtraction of 8-bit HEX Numbers

Test Case ID	Input / Scenario	Expected Result
TC01	12H + 34H	46H
TC02	50H - 20H	30H
TC03	FFH + 01H	Carry generated

Addition/Subtraction of 16-bit Numbers

Test Case ID	Input / Scenario	Expected Result
TC01	1234H + 1111H	2345H
TC02	2345H - 1234H	1111H
TC03	FFFFH + 0001H	Carry generated

Fibonacci Series

Test Case ID	Input / Scenario	Expected Result
TC01	n=5	0 1 1 2 3
TC02	n=1	0
TC03	n=10	Correct sequence generated

Odd and Even Series

Test Case ID	Input / Scenario	Expected Result
TC01	First 10 odd numbers	1,3,5...
TC02	First 10 even numbers	2,4,6...
TC03	Count check	10 values displayed

BCD Addition

Test Case ID	Input / Scenario	Expected Result
TC01	25 + 15	40
TC02	99 + 01	Carry adjustment
TC03	00 + 00	00

BCD Multiplication

Test Case ID	Input / Scenario	Expected Result
TC01	12 x 02	24
TC02	10 x 10	100
TC03	00 x 25	0

Division of Two 8-bit Numbers

Test Case ID	Input / Scenario	Expected Result
TC01	20 / 04	Quotient=5
TC02	15 / 02	Quotient and remainder
TC03	Division by zero	Error handling